

CO3_AT3_PART1
[MLA0305_reinforcement_learning](#)

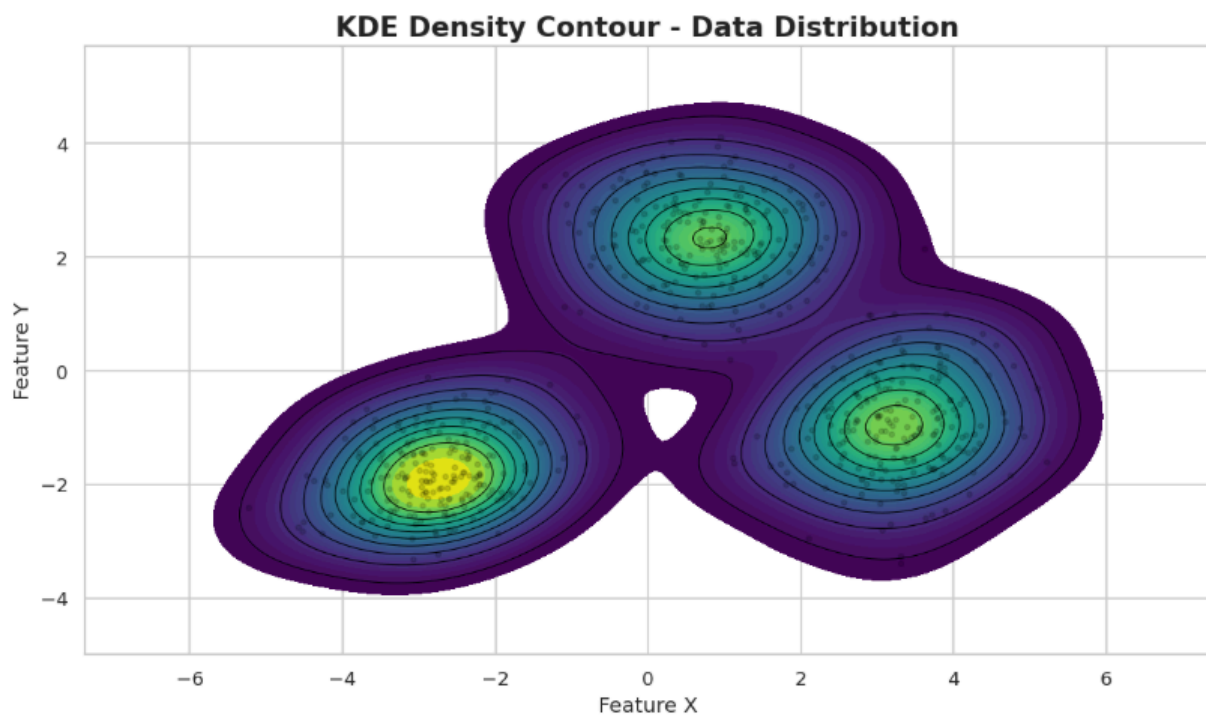
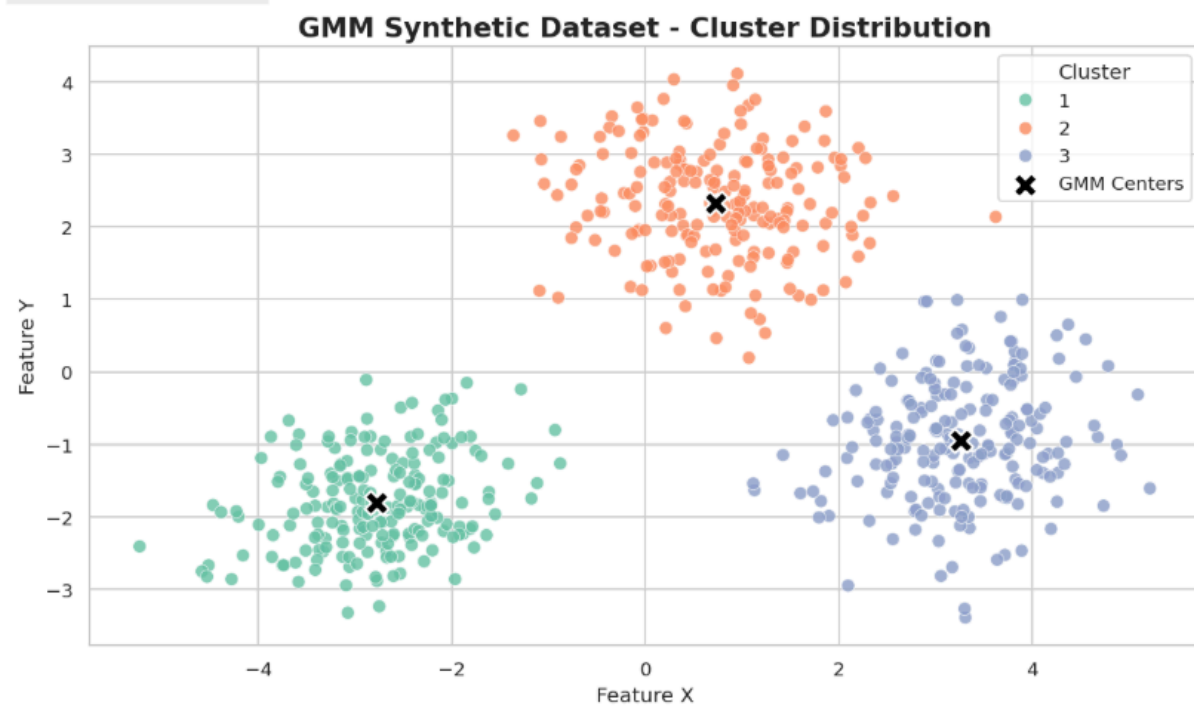
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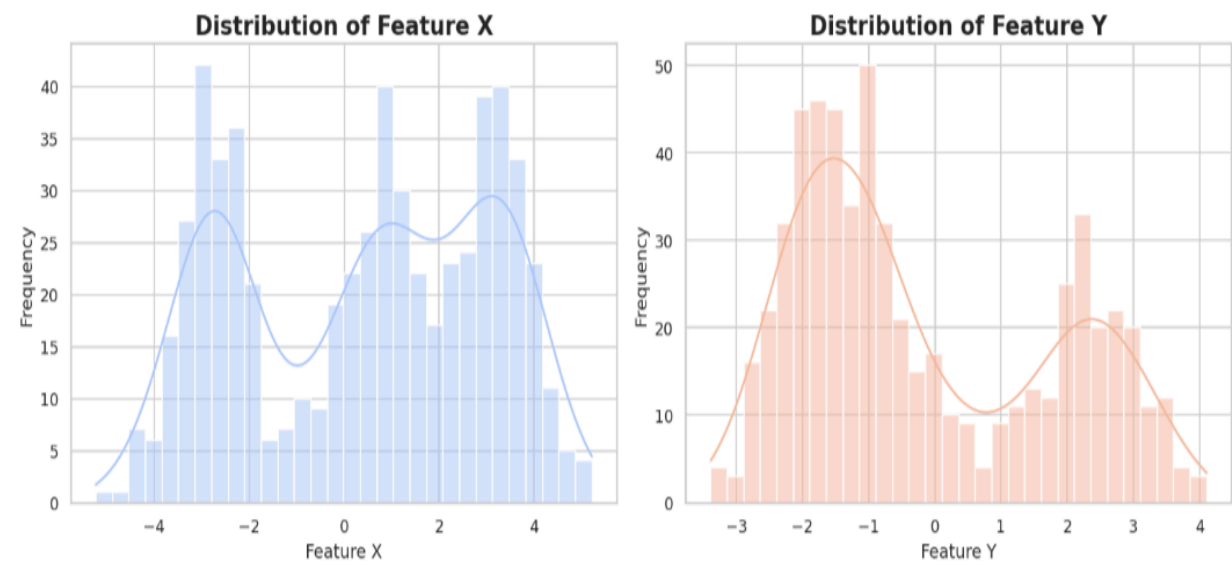
Vanilla Policy Gradient (REINFORCE) — Policy Optimization Performance Analysis

GMM Generated Synthetic Dataset			
	Feature_X	Feature_Y	Cluster
0	1.627	2.016	2
1	3.799	-0.824	3
2	3.057	-2.814	3
3	1.473	2.257	2
4	-2.214	-1.999	1
5	-2.437	-2.282	1
6	-2.084	-2.194	1
7	2.659	0.254	3
8	0.056	1.463	2
9	3.791	0.419	3

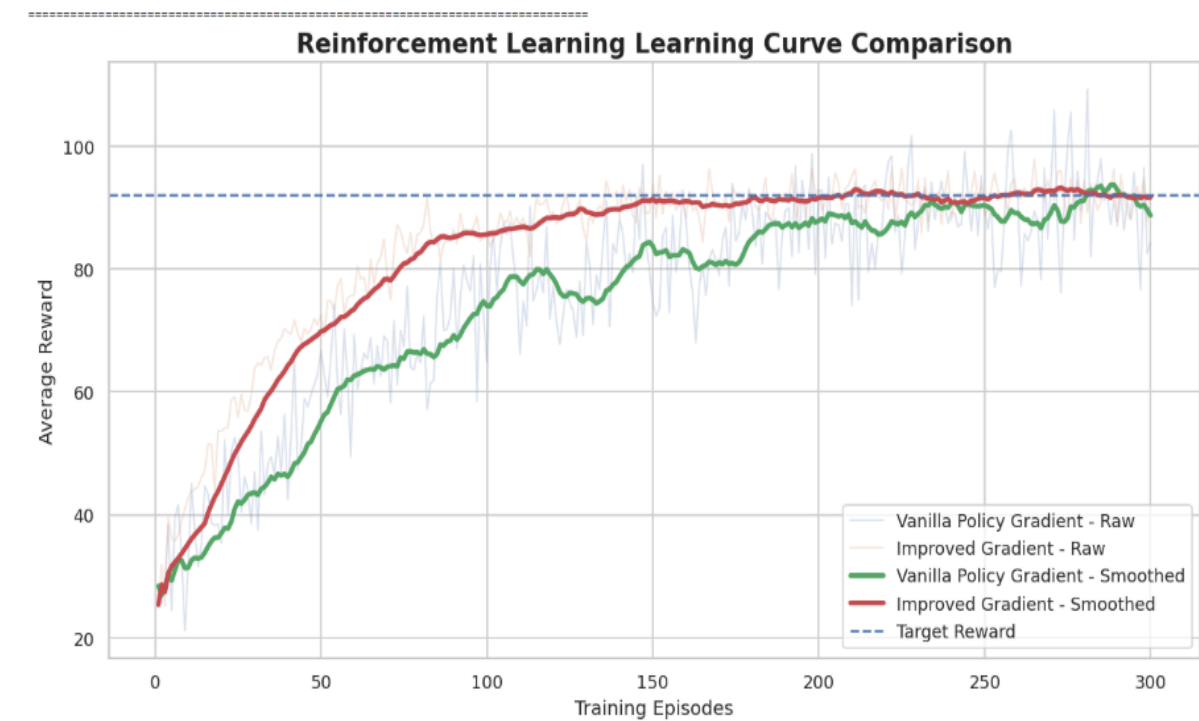
Fig1: - Sample Dataset



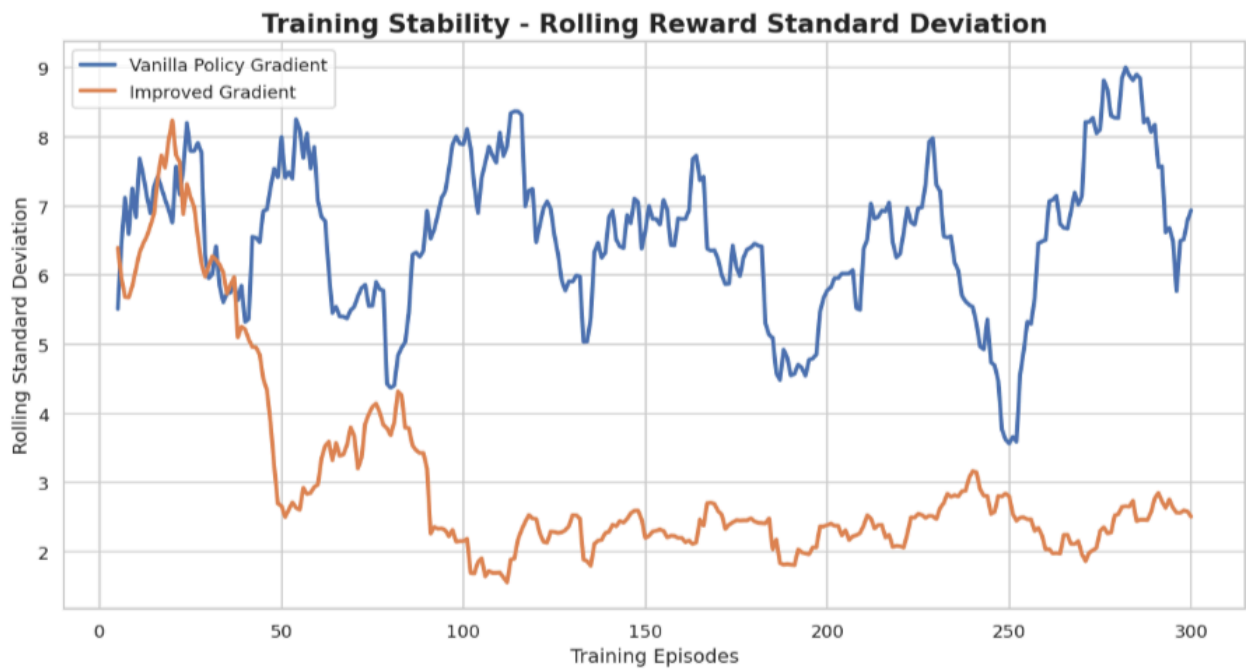
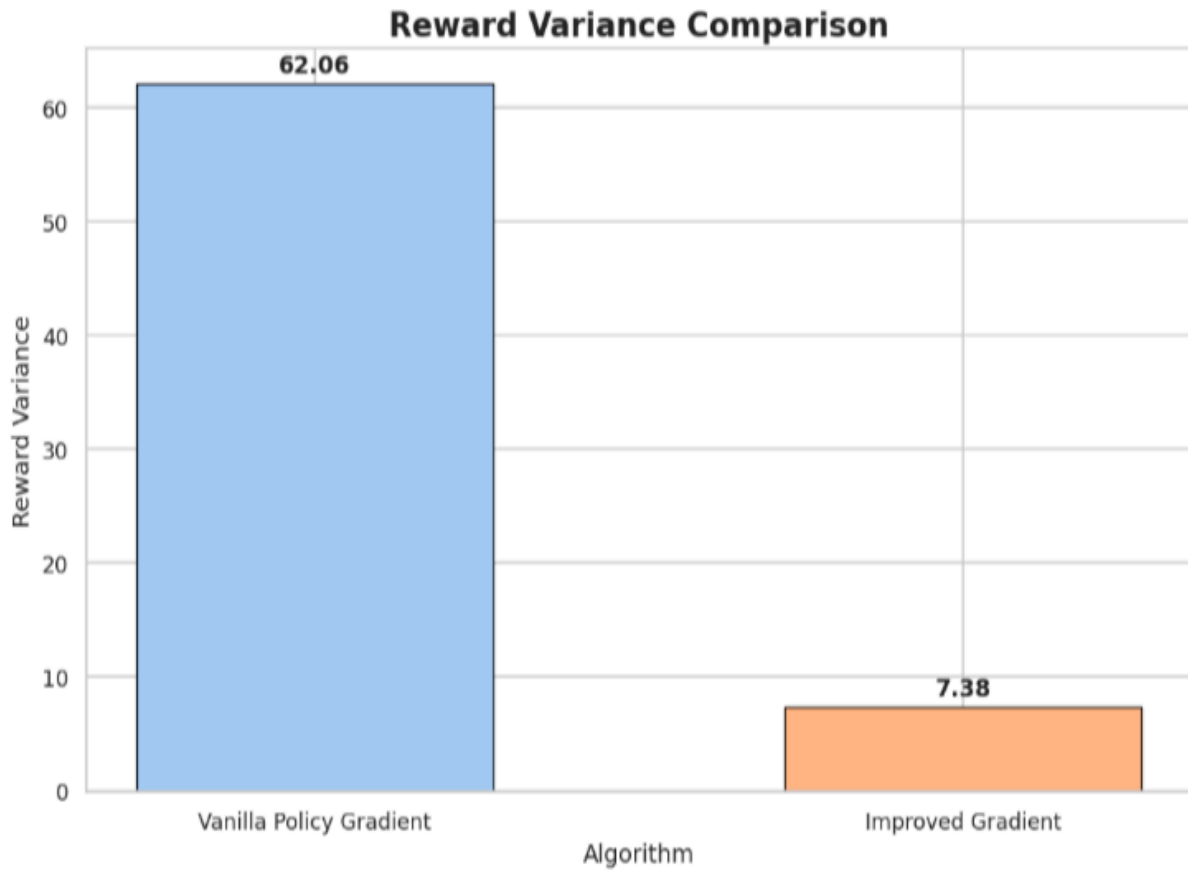
the three major primary features and report on the three features.



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Adding primary and lower-level goals.



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ALGORITHM COMPARISON SUMMARY

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Vanilla Policy Gradient vs Improved Gradient Method

	Algorithm Name	Variance	Stability	Learning Speed	Convergence
0	Vanilla Policy Gradient	62.060000	Low	Moderate	~147 episodes
1	Improved Gradient Method	7.380000	Very High	Fast	~81 episodes

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KEY PERFORMANCE RESULTS

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Reinforcement Learning Performance Metrics

	Metric	Result
0	Vanilla Reward Variance	62.06
1	Improved Reward Variance	7.38
2	Variance Reduction	88.1%
3	Vanilla Convergence	147 episodes
4	Improved Convergence	81 episodes
5	Convergence Speed Improvement	44.9%